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Assignments :-

Module – 1 : CS – Introduction

1. What is meaning of Cyber Security ?

Ans. Cyber Security is a protect the system form hacker like a computer, network and data from attack, damage or any unauthorize access, include firewall, antivirus, encryption(cryptographic), monitoring security to keep your system safe.

2. What are the main objectives of cyber security?

Ans. The main objectives of cybersecurity are to protecting information, systems, and networks from cyber threats. These objectives are using the CIA(Confidentiality Integrity Availability) Triad.

1. Confidentiality :-

- Protect sensitive data from unauthorized access and data breach.
- Uses tools like encryption, access controls, and authentication.

- Techniques include encryption, access controls, and authentication mechanisms.

2. Integrity :-

- Protecting information from being altered, tampered with, or destroyed in an unauthorized way.
- Integrity helps maintain the trustworthiness of data by having it in the correct state and immune to any improper modifications.
- Techniques include hashing, digital signatures, and checksums.

3. Availability :-

- The authorized users should be able to access data whenever required.
- Protects against disruptions such as cyberattacks (e.g., DDoS), hardware failures, or natural disasters.
- Techniques include redundancy, backups, and robust network architectures.

3. What is offensive and defensive in cyber security?

Ans. **1. Offensive :-** The Offensive security is the called Red Teaming. It is Ethical Hacking a system, network, or application to exploit and find vulnerabilities, software bugs, weakness point, etc, to protect the data and systems. It is the protect the system and network for hackers it called Offensive Cyber Security.

2. Defensive :- The Defensive security is the called Blue Teaming. It is protecting systems, networks, and data from cyberattacks. It focuses on the continuous monitoring, detection, and response to security threats. Prevent attacks and block malicious activity. It is called a Defensive Cyber Security.

4. What is cyberspace and law?

Ans. Cyberspace law refers to the legal framework governing the internet, digital activities, and online interactions. It encompasses laws, regulations, and policies addressing issues like a...

- Data protection and privacy
- Cybercrime and digital evidence
- Intellectual property rights
- Online contracts and e-commerce
- Cyber security and hacking
- Digital forensics and electronic evidence
- Internet governance and jurisdiction

5. What is cyber welfare?

Ans. Cyber welfare generally refers to social-protection, initiatives, measures, or policies aimed at promoting digital well-being, safety, and security for individuals and communities online. It encompasses aspects like a...

- Protecting users from cyber threats (e.g., phishing, scams, malware).
- Promoting digital literacy and responsible online behavior.
- Supporting victims of cybercrime.
- Ensuring accessibility and inclusivity in cyberspace.
- Encouraging healthy digital habits and screen time management.

6. Explain the Types of Hacker?

Ans. There are a Three types based of hackers.

1. White Hat Hackers :-

- Work legally with government permissions.
- Help organizations find security vulnerabilities.
- Improve systems by performing penetration testing.
- Aim to protect data, not steal it.
- Ex:- A cybersecurity professional hired to test a bank's network

2. Black Hat Hackers :-

- Hack illegally without any permissions.
- Steal data, damage systems, or plant malware.
- Motivated by money, revenge, or personal gain.
- Responsible for cybercrimes like ransomware attacks.
- Ex:- Someone breaking into a company's server to steal credit cards.

3. Grey Hat Hackers :-

- Operate between ethical and unethical behavior.
- Hack without any permissions, but usually without malicious intent.
- May report vulnerabilities but often after exploiting or testing them.
- Can cause harm unintentionally.
- Ex:- A hacker who breaks into a system, finds a flaw, and then informs the company but still acted illegally.

7. What is the full form of SOC in cyber security?

Ans. The full form of SOC is a Security Operations Center.

8. What are the Challenges of Cyber Security?

Ans. The Cybersecurity evolving challenges due to advancing technology, increasing connectivity, and the sophistication of attackers.

1. Sophistication of cyber attacks :-

- Attackers now use advanced tools such as AI malware, viruses, and zero-day exploits.

2. Rapidly Expanding Attack Surface :-

- Growth of cloud computing, remote work, IoT devices, and mobile technologies creates more entry points for attackers.

3. Shortage of Skilled Cybersecurity Professionals :-

- There is a global talent gap, making it hard for organizations to defend their systems effectively.

4. Human Error :-

- Human accidentally clicking malicious links, using weak passwords, or misconfiguring systems remains one of the top causes of breaches.

5. Ransomware and Financially Attacks :-

- Ransomware incidents are increasing, targeting hospitals, governments, and businesses.

6. AI and Automation used by Attackers :-

- Attackers use AI to automate attacks, generate phishing messages, and bypass detection systems.

7. Privacy and Data Protection Issues :-

- Organizations handle vast amounts of sensitive data (personal, financial, health) making them prime targets.